

CRUD Section – Explanation of SQL Commands and Final Improvements

In this section, I presented the four fundamental SQL operations—**Create, Read, Update, and Delete**—which I applied to manage data in the Members table of the gym database. I initially created four screenshots to demonstrate each of these commands in action, and later decided to make some improvements after reviewing the quality and clarity of the original work. The last six screenshots in this section reflect those enhancements.

1. Create (Insert a New Member)

In the first screenshot, I used the `INSERT INTO` command to add a new member named **George Miller** to the Members table. I included all the required fields, such as his name, contact details, and the membership start and end dates. The successful insertion was confirmed by the message *"1 row(s) affected"* in the output.

2. Read (View All Members)

The second screenshot shows the `SELECT * FROM Members;` command, which I used to view all the existing records in the table. This allowed me to confirm that George Miller had been successfully added and was listed among the current members.

3. Update (Edit a Member's Phone Number)

In the third screenshot, I demonstrated how to update George Miller's phone number using the `UPDATE` command. I used his MemberID (17) to target the correct entry. Again, the confirmation *"1 row(s) affected"* in the output showed that the update had been completed successfully.

4. Delete (Remove Duplicate Entry)

The fourth screenshot highlights the `DELETE` command I used to remove a duplicate or incomplete record of George Miller, identified by `MemberID = 24`. The deletion was also confirmed with the message *"1 row(s) affected"*.

Final Improvements – Last Six Screenshots

After completing the initial four CRUD operations, I reviewed my work and decided to improve the overall presentation and realism of the data. As a result, I added six new screenshots to demonstrate these improvements.

I realised that I had hardcoded some values—such as MemberID—which isn't ideal in real-world scenarios, especially when the database is set to auto-increment primary keys. To

address this, I created a new insertion example with a different member, **Bob Jones**, and allowed the system to handle the ID generation automatically.

I also took care to make the data more consistent and professional. For example, I used a new email address to avoid the issue of inserting duplicate values, which could potentially break the system depending on how constraints are defined.

These updated screenshots also show clearer formatting and better structure in the SQL queries, making them easier to follow and more in line with best practices. The result outputs in each case confirmed that the operations were still functioning correctly, with messages like *"1 row(s) affected"* displayed clearly.

Overall, I made these final additions to improve the realism, clarity, and technical quality of my work. They reflect not just the basic functionality of CRUD operations, but also my attention to detail and willingness to refine my project to a higher standard.

The screenshot displays a SQL IDE interface. The top toolbar includes icons for file operations and a status bar at the bottom right with 'Apply' and 'Revert' buttons. The main editor shows a SQL query with four lines: 1. `USE gymdb;`, 2. `INSERT INTO Members (FirstName, LastName, DOB, Email, Phone, Address, MembershipID, StartDate, EndDate)`, 3. `VALUES ('George', 'Miller', '1991-07-20', 'george.miller2@email.com', '555-00600', '6 Main St', 1, '2025-07-01', '2026-07-01');`, and 4. A blank line. The 'Action Output' pane at the bottom shows a log of database actions and their results.

	Time	Action	Response	Duration / Fetch Time
✓ 1	17:53:39	CREATE DATABASE gymdb	1 row(s) affected	0.012 sec
✓ 2	17:53:39	USE gymdb	0 row(s) affected	0.00058 sec
✓ 3	17:57:48	USE gymdb	0 row(s) affected	0.018 sec
✓ 4	17:57:48	CREATE TABLE IF NOT EXISTS Memberships (MembershipID INT PRIMARY KEY AUTO_INCREMENT, Type VARCHAR(50) NOT NULL)	0 row(s) affected	0.030 sec
✓ 5	17:57:48	CREATE TABLE IF NOT EXISTS Members (MemberID INT PRIMARY KEY AUTO_INCREMENT, FirstName VARCHAR(50) NOT NULL, LastName VARCHAR(50) NOT NULL, DOB DATE, Email VARCHAR(100) NOT NULL, Phone VARCHAR(20) NOT NULL, Address VARCHAR(255) NOT NULL, MembershipID INT NOT NULL)	0 row(s) affected	0.0064 sec
✓ 6	17:57:48	CREATE TABLE IF NOT EXISTS Trainers (TrainerID INT PRIMARY KEY AUTO_INCREMENT, FirstName VARCHAR(50) NOT NULL, LastName VARCHAR(50) NOT NULL, DOB DATE, Email VARCHAR(100) NOT NULL, Phone VARCHAR(20) NOT NULL, Address VARCHAR(255) NOT NULL, Salary DECIMAL(10,2) NOT NULL)	0 row(s) affected	0.0073 sec
✓ 7	17:57:48	CREATE TABLE IF NOT EXISTS Classes (ClassID INT PRIMARY KEY AUTO_INCREMENT, Name VARCHAR(100) NOT NULL, Description VARCHAR(255) NOT NULL, ScheduleID INT NOT NULL, ClassTime TIME NOT NULL, Capacity INT NOT NULL)	0 row(s) affected	0.0032 sec
✓ 8	17:57:48	CREATE TABLE IF NOT EXISTS ClassSchedule (ScheduleID INT PRIMARY KEY AUTO_INCREMENT, ClassID INT NOT NULL, MemberID INT NOT NULL, StartDate DATE NOT NULL, EndDate DATE NOT NULL, Status VARCHAR(20) NOT NULL)	0 row(s) affected	0.0053 sec
✓ 9	17:57:48	CREATE TABLE IF NOT EXISTS Bookings (BookingID INT PRIMARY KEY AUTO_INCREMENT, MemberID INT NOT NULL, ClassID INT NOT NULL, StartDate DATE NOT NULL, EndDate DATE NOT NULL, Status VARCHAR(20) NOT NULL)	0 row(s) affected	0.0046 sec
✓ 10	17:57:48	CREATE TABLE IF NOT EXISTS Payments (PaymentID INT PRIMARY KEY AUTO_INCREMENT, MemberID INT NOT NULL, Amount DECIMAL(10,2) NOT NULL, PaymentDate DATE NOT NULL, Status VARCHAR(20) NOT NULL)	0 row(s) affected	0.0042 sec

walford/Desktop/01_create_member.png.sql

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Query 1

gym_all_queries*

gym_database*

01_create_member.png*

SQL File 10

Limit to 1000 rows

1 •

USE gymdb;

2 •

-- 02_read_members.png

3 •

SELECT * FROM Members;

100%

23:3

Result Grid

Filter Rows: Search

Edit:

Export/Import:

Object Info

Session

No object selected

MemberID	FirstName	LastName	DOB	Email	Phone	Address	MembershipID	StartDate	EndDate
1	Alice	Smith	1990-01-01	user1@email.com	555-99999	1 Main St	1	2024-01-01	2025-01-01
2	Bob	Brown	1992-09-27	user2@email.com	555-00200	2 Main St	2	2024-01-31	2025-01-31
3	Charlie	Johnson	1995-06-24	user3@email.com	555-00300	3 Main St	1	2024-03-01	2025-03-02
4	Diana	Williams	1998-03-20	user4@email.com	555-00400	4 Main St	2	2024-03-31	2025-04-01
5	Ethan	Davis	2000-12-14	user5@email.com	555-00500	5 Main St	1	2024-04-30	2025-05-01
17	George	Miller	1991-07-20	george.miller@email.com	555-00600	6 Main St	1	2025-07-01	2026-07-01
HULL	HULL	HULL	HULL	HULL	HULL	HULL	HULL	HULL	HULL

1 •

USE gymdb;

2 •

UPDATE Members

3 •

SET Phone = '555-99999'

4 •

WHERE MemberID = 17;

5 •

100%

1:5

Action Output

	Time	Action	Response	Duration / Fetch Time
✓ 1	17:53:39	CREATE DATABASE gymdb	1 row(s) affected	0.012 sec
✓ 2	17:53:39	USE gymdb	0 row(s) affected	0.00058 sec
✓ 3	17:57:48	USE gymdb	0 row(s) affected	0.018 sec
✓ 4	17:57:48	CREATE TABLE IF NOT EXISTS Memberships (MembershipID INT PRIMARY KEY AUTO_INCREMENT, Type VARCHAR(50) N...	0 row(s) affected	0.030 sec
✓ 5	17:57:48	CREATE TABLE IF NOT EXISTS Members (MemberID INT PRIMARY KEY AUTO_INCREMENT, FirstName VARCHAR(50) N...	0 row(s) affected	0.0064 sec

[illegible]

Local instance 3306 - Warning - not supported

AdministrationSchemasQuery 1gym_all_queries*gym_database*01_create_member.png*SECTION_4_Full_Journey_Sample_Data*

MANAGEMENT

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Object Info

Session

No object selected

Limit to 1000 rows

12345678910

1USE gymdb;

2INSERT INTO Members (

3 FirstName, LastName, DOB, Email, Phone, Address, MembershipID, StartDate, EndDate

4) VALUES (

5 'Alice', 'Smith', '1990-05-10', 'alice.smith@email.com', '555-1234',

6 '123 Maple St', 2, '2025-06-01', '2026-06-01'

7);

8

9SELECT LAST_INSERT_ID();

10

100%1:10

Result GridFilter Rows: SearchExport:

LAST_INSERT_ID()
1

Result 2Read Only

Query Completed

O ChatGPT pode cometer erros. Considere verificar informações importantes.

MySQL Workbench

Local instance 3306 - Warning - not supported

AdministrationSchemasQuery 1gym_all_queries*gym_database*01_create_member.pngSECTION_4_Full_Journey_Sample_Data*

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PERFORMANCE

- Object Info
- Session

No object selected

Limit to 1000 rows

1USE gymdb;

2INSERT INTO Members (

3 FirstName, LastName, DOB, Email, Phone, Address, MembershipID, StartDate, EndDate

4) VALUES (

5 'Bob', 'Jones', '1988-02-12', 'alice.smith@email.com', '555-5678',

6 '456 Oak Ave', 1, '2025-07-01', '2026-07-01'

7);

8

100%1:8

Apply

Action Output

	Time	Action	Response	Duration / Fetch Time
✓ 24	16:23:45	USE gymdb	0 row(s) affected	0.00020 sec
✓ 25	16:23:45	CREATE TABLE Members (MemberID INT...	0 row(s) affected	0.022 sec
✓ 26	16:27:17	USE gymdb	0 row(s) affected	0.017 sec
✓ 27	16:27:17	INSERT INTO Members (FirstName, LastN...	1 row(s) affected	0.019 sec
✓ 28	16:27:17	SELECT LAST_INSERT_ID() LIMIT 0, 1000	1 row(s) returned	0.0014 sec / 0.00000...
✓ 29	16:29:43	USE gymdb	0 row(s) affected	0.019 sec
✗ 30	16:29:43	INSERT INTO Members (FirstName, LastN...	Error Code: 1062. Duplicate entry 'alice.smith@email....	0.016 sec

Query interrupted

MySQL Workbench

Local instance 3306 - Warning - not supported

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Object Info

Session

No object selected

Query 1

gym_all_queries*

gym_database*

01_create_member.png*

SECTION_4_Full_Journey_Sample_Data*

Limit to 1000 rows

1

2

3

USE gymdb;

SELECT * FROM Members;

100%

1:3

Result Grid

Filter Rows:

Search

Edit:

Export/Import:

MemberID	FirstName	LastName	DOB	Email	Phone	Address	MembershipID	StartDate	EndDate
1	Alice	Smith	1990-05-10	alice.smith@email.com	555-1234	123 Maple St	2	2025-06-01	2026-06-01

Members 1

Apply

Revert

Action Output

	Time	Action	Response	Duration / Fetch Time
1	16:32:49	USE gymdb	0 row(s) affected	0.0012 sec
2	16:32:49	SELECT * FROM Members LIMIT 0, 1000	1 row(s) returned	0.0017 sec / 0.00002...

Query Completed

MySQL Workbench

Local instance 3306 - Warning - not supported

Administration

Schemas

Query 1

gym_all_queries*

gym_database*

01_create_member.png*

SECTION_4_Full_Journey_Sample_Data*

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Object Info

Session

No object selected

1 • USE gymdb;

2 • UPDATE Members

3 SET Phone = '555-9999'

4 WHERE Email = 'alice.smith@email.com';

5

100%

1:5

Action Output

	Time	Action	Response	Duration / Fetch Time
3	16:34:12	USE gymdb	0 row(s) affected	0.022 sec
4	16:34:12	UPDATE Members SET Phone = '555-9999'...	1 row(s) affected Rows matched: 1 Changed: 1 Warni...	0.022 sec

Query Completed

MySQL Workbench

Local instance 3306 - Warning - not supported

Administration

Schemas

Query 1

gym_all_queries*

gym_database*

01_create_member.png*

SECTION_4_Full_Journey_Sample_Data*

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Object Info

Session

No object selected

Limit to 1000 rows

1

•

USE gymdb;

2

•

DELETE FROM Members

3

WHERE Email = 'alice.smith@email.com';

4

100%

1:4

Action Output

	Time	Action	Response	Duration / Fetch Time
✓ 5	16:36:44	USE gymdb	0 row(s) affected	0.017 sec
✓ 6	16:36:44	DELETE FROM Members WHERE Email = 'alic...	1 row(s) affected	0.014 sec

Query Completed